MIL-S-51368B(EA)
26 June 1979
SUPERSEDING
MIL-S-51368A(EA)
20 January 1976

MILITARY SPECIFICATION

SHELTER SYSTEM, COLLECTIVE PROTECTION, CHEMICAL-BIOLOGICAL : INFLATABLE, 10-MAN, TRAILER TRANSPORTED, M51

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers collective protection equipment consisting of a shelter, entrance, filter and utilities unit, and accessory package.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	-	Steel Strapping, Flat and Seals.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-c-843	-	Cushioning Material, Cellulosic.
PPP-C-l120	-	Cushioning Material, Uncompressed Bound Fiber for Packaging.
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

FSC 4240

: Beneficial comments (recommendations, additions, deletions) and any : pertinent data which may be of use in improving this document should : be addressed to: Commander, US Army Armament Research and Development : Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by : wing the self-addressed Standardization Document Improvement Proposal; : (DD Form 1426) appearing at the end of this document or by letter. :

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:58Z
Check the source to verify that this is the currentversion before use.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-F-51366 - Filters, Particulate: Recirculation (550 CFM and 1200 CFM).
MIL-F-51369 - Filters, Gas: Recirculation (550 CFM and 1200 CFM)
MIL-S-51422 - Shelter and Entrance Assembly - Chemical-Biological Shelter System: Inflatable.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-281 - Automobile, Trucks, Truck-Tractor, Trailers and Trailer Dollies, Preservation and Packaging of.
MIL-STD-781 - Reliability Tests: Exponential Distribution.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

E5-19-6523 - Shelter System Collective Protection, Chemical-Biological: Inflatable, 10-Man, Trailer Transported, M51.
D150-2-1 - Marking Diagram for Shipping Containers.
DL5-19-6821 - M68 Filters and Utilities Unit with Accessories,
DL5-19-6822 - Shelter and Entrance Assembly with Accessories.

PUBLICATIONS

CHEMICAL SYSTEMS LABORATORY TECHNICAL MANUAL

TM3-4240-264-12 - Shelter System, Collective Protection, Chemical-Biological: M51 (Operating and Organizational Maintenance).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:58Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF HEATING, REFRIGERATIOG, AND AIR CONDITIONING ENGINEERS, INC.

Guide and Data Book - NCA-60 Curve, Noise Criterion.

(Application for copies should be addressed to the American Society of Heating, Refrigeration, and Air Conditioning Engineers, Inc. , 345 East 47th Street, New York, New York 10017).

3. REQUIREMENTS

3.1 Materials and components. The M51 shelter system can be procured as a complete system as described by Drawing E5-19-6523, or as two subsystems to be consolidated in depot. The two subsystems are described by Drawings PL5-19-6821 (M68 Filters and Utilities Unit with Accessories) and PL5-19-6822 (Shelter and Entrance Assembly with Accessories)*

3.1.1 Material. All materials cited on Drawing E5-19-6523, PL5-19-6821, or PL5-19-6822 specified by the contract or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings,

3.1.2 Components. All components of the shelter system shall conform to the specifications and drawings listed on Drawing E5-19-6523, PL5-19-6821, or PL5-19-6822 as specified by the contract and their subsidiary drawings,

3.2 Manufacture and assembly. The M51 shelter system shall be manufacture and assembled as specified on Drawing E5-19-6523, PL5-19-6821. or PL5-19-6822 as specified by the contract.

3.3 Performance.

3.3.1 Initial operating mode. *When* the M51 shelter system is fully erected, the heater and the air conditioner shall be operated for 2.0 hours each in their respective modes. The operation in each mode need not be continuous. The neater and the air conditioner shall cycle automatically when tested in accordance with 4.4.4.1.

3.3.2 Operating pressures. A pressure of 1.0 (+0,1 or -0.2) inches water guage (iwg) shall be maintained in the shelter and a pressure of 0.5 (+0.1 or -0.2) iwg shall be maintained in the entrance during recirculation mode operation except for during entry and exit periods,

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:58Z
Check the source to verify that this is the currentversion before use.

The shelter shall maintain a positive pressure in all modes, i.e. recirculation, cooling, and heating at all times. During an entry or exit, the shelter low pressure alarm shall not activate when tested in accordance with 4.4.4.1. During operation in the heating and cooling modes (without entry or exit) the shelter and entrance pressures shall fall within the limits specified in table I.

3.3.3 Entry and exits. A minimum of twelve entry and exits shall be made during the four hour operating mode. When tested in accordance with 4.4.4.1 pressures in the shelter and entrance shall stabilize to the values specified in 3.3.2 after an entry or exit. NOTE: An entry or exit is defined as going from the outside through the entrance into the shelter or going from the shelter through the entrance to the outside.

3.3.4 System leakage. With 160 cubic feet per minute (cfm) maximum of make-up air being introduced into the system and with the shelter and entrance maintaining at least minimum pressures (3.3.2), the purge-air effluent, measured at the damper valve in the outside entrance door shall be no less than 110 cm when tested in accordance with 4.4.4.2.

3.3.5 Entrance purge. The entrance shall be capable of purging dioctylphthalate (DOP) to give a 1000 to 1 (3 log) reduction in 5.0 minutes maximum when tested in accordance with 4.3.3.1.

3.3.6 Shelter purge. The shelter shell be capable of purging DOP to give a 1000 to 1 (3 log) reduction in 60 minutes maximum when tested in accordance with 4.3,3.1.

3.3.7 Noise. Noise inside the shelter shall be determined for vent, cool and heat modes using applicable instrumentation and shall not exceed the noise criterion curve NCA-60 of the ASHRAE Guide and Data Book when tested in accordance with 4.3.3.2.

3.4 Reliability.

3.4.1 Preproduction and reliability qualification. Prior to the start of regular production, a preproduction sample of the M51 shelter system shall be produced in accordance with this specification for examination (4.3.2.1), test (4.3.2.2), and reliability qualification (demonstration) phase testing (4.3.2.3).

3.4.1.1 Reliability qualification (demonstration) phase. The M51 shelter system shall conform to the following Mean Time Between Failure criteria, MTBF(θ_o), for both the overall system reliability and protection reliability when tested as specified in 4.4.4.3 and evaluated for accountable failures (3.4.3) in accordance with table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:58Z
Check the source to verify that this is the current version before use.

TABLE I - Reliability performance requirements

Requirement	Operating parmeters
Shelter	
1. Temperature*	60°F to 90°F
2. Airflow to shelter	Minimum mean: 900 cfm Minimum reading: 850 cfm-I
3. Static pressure	Vent mode set point: 1(+0.1 or -0.2) iwg Minimum value all modes: mask alarm sound Maximum value all modes: 2.0 iwg
Airlock	
1. Airflow thru filter	Minimum mean: 375 cfm Minimum reading: 300 cfm
2. Static pressure	Vent mode set point: 0.5 (+0.1 or-O.2) iwg Minimum value all modes: 0,3 iwg
Filter and utilities unit	
1. Make-up airflow rate	Minimum mean: 140 scfm Minimum reading: 135 scfm
2. Inflation air pressure	45 iwg maximum 30 iwg minimum

* Automatically controlled after manual mode selection.

MTBF Criteria:

(a) Overall system reliability - The overall system reliability for the M51 shall be no less than a specified MTBF (θ_o) of 342 hours as evaluated in accordance with test plan VII of MIL-STD-781.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:58Z
Check the source to verify that this is the current version before use.

(b) Protection reliability - The protection reliability for the M51 shall be no less than a specified MTBF (Θ_o) of 705 hours as evaluated in accordance with test plan VII of MIL-STD-781,

3.4.2 Reliability production acceptance (sampling) phase. The production acceptance sample of M51 shelter systems shall demonstrate an overall system and protection reliability of 0.90 and 0.95 respectively when tested as specified in 4.4.4.4 and evaluated in accordance with 4.4.3.3.

3.4.3 Accountable failure definition. Deviations from table I, reliability performance requirements, shall be considered as accountable failures ; either a protection failure or an overall system failure. however, in procurement of separate subsystems (per PL5-19-6821 or PL5-19-6822), the failure of equipment provided as "Government furnished equipment" shall not constitute an accountable failure but shall be considered a malfunction of test support equipment.

3.4.3.1 Protection failure. An accountable failure of protection is defined as occurring when the alarm indicates low pressure in the shelter for a period of 30 seconds, thereby causing the shelter occupants to don protective masks. During reliability qualification testing, if *any* component shows indications of a possible protection failure from either visible or audible symptoms, that component may not be replaced during that 24 hour period.

3.4.3.2 Overall system failure. An accountable failure of the overall system is defined as an occurrence of a malfunction, component failure or loss of system operation not caused by operator error, or test equipment or facility failure under any of the following conditions :

(1) Repair and/or replacement of a component requiring time in excess of 30 minutes to repair.

(2) Total maintenance time per 24 hour test cycle greater than 1 hour.

(3) Repair and/or replacement of any one component more often than every 1,000 hours

(i.e. $\frac{\text{Total test hours}}{\text{Number of repairs or replacements}} < 1{,}000$).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:58Z
Check the source to verify that this is the current version before use.

3.5 Workmanship. All M51 shelter system equipment, not otherwise covered by applicable specifications, shall be free from damage (cracks, breaks , warpage, sharp edges, or fabric tears) and foreign matter (dirt, grease, or oil).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.1 Objective evidence. The contractor shall provide objective evidence acceptable to tie contracting officer that the requirements of 3.1 and section 5, for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Preproduction and reliability qualification (demonstration) phase (see 4.3)。

b. Quality conformance inspection (4. 4)

4.3 Preproduction and reliability qualification (demonstration) phase.

4.3.l Sample. A sample of four M51 shelter systems shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production. When the system is procured in two, separate subsystems (PL5-19-6821 and PL5-19-6822), the sample size shall be four of each subsystem.

4.3.2 Inspection procedure (ref. table 11).

4.3.2.1 For examination. Each system in the sample of four M51 shelter systems shall be examined for the defects listed in 4.4.3.4 in conjunction with the applicable drawings and their specifications.

4.3.2.2 For tests. Each system in the sample of four M51 shelter systems shall be tested in accordance with 4.3.3.1, 4.4.4.1 and 4.4.4.2. One system shall be tested in accordance with 4.3.3.2,
(NOTE: These tests shall be performed prior to testing in accordance with 4.3.2.3.)

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:58Z
Check the source to verify that this is the currentversion before use.

Table II. Inspection schedule.

Inspection sequence	Type of inspection: Preproduction	Type of inspection: Quality conformance	Examination or test	Requirement paragraph	Inspection paragraph
1	x	x	Examination	-	4.4.3.4
2	x	x	Operating mode	3.3.1	4.4.4.1
3	x	x	Operating pressures	3.3.2	4.4.4.1
4	x	x	Entries and exits	3.3.3	4.4.4.1
5	x	x	System leakage	3.3.4	4.4.4.2
6	x	-	Entrance purge	3.3.5	4.3.3.1
7	x	-	Shelter purge	3.3.6	4.3.3.1
8 & 1	x	-	Noise	3.3.7	4.3.3.2
9	x	-	Reliability (Pre-Pro)	3.4.1	4.4.4.3
10	-	x	Reliability(Production)	3.4.2	4.4.4.4

4.3.2.3 Reliability qualification (demonstration) phase. The sample of four M51 shelter systems shall be simultaneously tested, in 24 hour periods, as indicated below, and in accordance with 4.4.4.3. The reliability test shall be run on a 24 hour basis. The first 2 hours shall be run in the recirculation mode. The next 20 hours shall consist of alternate 1 hour heat and 4 hour cooling cycles giving a total of four each heating and cooling cycles per 24 hours. Thirty minutes after the initiation of each recirculation, heating, and cooling cycle the operating parameters of table I shall be measured and recorded. The remaining 2 hours of the 24 hour period are for striking and erecting the units. A maximum of 1 hour maintenance is allowed during each 24 hour periods Every effort shall be made to have each unit accumulate the same number of hours. Where subsystems are procured separately, the testing of M68 Filters and Utilities Units with Accessories (PL5-19-6821) shall be conducted as specified above using Government furnished shelters and entrances. The subsystem defined by PL5-19-6822 (Shelter and Entrance Assembly with Accessories) shall be tested on a cyclic basis according to table III. A cycle is defined as one erection, one hour of operation with pressures recorded in the recirculation rode, and one strike. In the strike operation, the shelter and entrance assembly shall be prepared for storage but will not be stowed. The number of cycles shall be distributed among the shelter and entrance assemblies as nearly equal as practical. The following pressures shall be monitored during operation: shelter static pressure, airlock static pressure, and inflation air pressure. The M68 Filters and Utilities Unit shall be monitored to verify normal operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:58Z
Check the source to verify that this is the current version before use.

TABLE III . Preproduction reliability test

Failures	Reject	Accept
0	0	68 cycles/ 4 units
1	0	94 cycles/ 4 units
2	0	119 cycles/ 4 units
3	Less than 26 cycles/ 4 units	145 cycles/ 4 units
4	Less than 52 cycles/ 4 units	145 cycles/ 4 units
5	Less than 78 cycles/ 4 units	145 cycles/ 4 units
6	Less than 145 cycles/4 units	Not applicable

4.3.3 Tests. All tests listed in table II shall be performed when the unit is procured as a complete system. When the system is procured in separate subsystem, the M68 Filters and Utilities Unit and Accessories (PL5-19-6821) shall be tested to sequence numbers 1,2,3,4,5,6,7,8 and 9 in preproduction and to 1,2,3,4,5 and 10 in production (Quality Conformance). The Shelter and Entrance Assembly and Accessories (PL5-19-6822) shall be tested to sequence numbers 1,2,3,4,5 and 9 in preproduction and to 1,2,3,4,5, and 10 in production (Quality Conformance).

4,3.3.1 Entrance and shelter purge. A suitable DOP generator shall be placed in the volume to be purged. A sufficient concentration of DOP smoke shall be disseminated while the system is de-energized: The DOP generator shall then be turned off and the DOP detector shall be set at 100 percent. The system shall then be energized. This point in time is time zero (To). The M51 recirculation system shall purge the respective volumes in the times indicated in 3.3.5 and 3.3.6 respectively. NOTE : These tests shall be performed separately. Mixing fans should be placed on the floor to assure thorough mixing before initiating tests. These fans shall be disconnected at time zero (To).

4.3.3.2 Noise. Noise shall be determined inside the shelter at the geometric center at a height of 5 feet from the floor. This test shall be conducted in vent, cool and heat modes and shall meet the requirements of 3.3.7,

4.3.4 Acceptance/rejection criteria. Each shelter system in the preproduction sample shall meet the examination specified in 4.3.2.1 and tests specified in 4.3.2.2. Reliability qualification (4.3.2.3) shall be evaluated in accordance with test plan VII of MIL-STD-781. The minimum, expected, and maximum number of test hours to reach a decision are 1500, 2400, and 3200 hours respectively, The supplier shall obtain written approval from the contracting officer before proceeding with regular production,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:58Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of one M51 shelter system produced by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions as in the production lot. Any change in equipment or procedure used in regular production which differs from that used in the preproduction sample shall be approved by the Government prior to its institution.

4.4.2 Sampling.

4.4.2.1 For examination and non-destructive tests. Each shelter system shall be examined and tested. .

4.4.2.2 For reliability preduction acceptance phase. One sample shelter system shall be selected from each two months production and tested in accordance with 4.4.3*3 for those months when the production rate is 15 systems or greater. For those months when the production rate is less than 15, one system per even 30 accumulated systems shall be tested in , accordance with 4.4.3.3.

4.4.3 Inspection procedures (reference table II).

4.4.3.1 For examination. Each shelter system shall be examined in accordance with the list of defects (4.4.3.4); any defect shall reject the system.

4.4,3.2 For test. Each shelter system shall be tested as specified in 4.4.4.1 4.4.4.2. Failure of any test shall reject the system.

4.4.3.3 For production reliability. The M51 shelter system shall be . tested as specified in 4.4.4.4. The table IV lists reliability requirements for the complete system and for separate subsystems (tested with Government furnished components being used" as "test equipment"). During testing of the Shelter and Entrance Assembly with Accessories (PL5-19-6822) only the following pressures shall be monitored: shelter static pressure, airlock static pressure, and inflation air pressure. Failure to meet the criteria listed below shall be cause of rejection of the production quantity until the cause for failure has been corrected and the correction verified. The Government may require a retest to verify that the cause of the failure has been corrected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:58Z
Check the source to verify that this is the current version before use.

TABLE IV. Production reliability

	Complete System		Shelter & Entrance Assy with Accessories		M68 Filters & Utilities Unit with Accessories	
	Failure	Cycles	Failures	Cycles	Failures	Cycles
FOR SYSTEM RELIABILITY	0	9	0	11	0	11
	1	19	1	23	1	23
	Failures	Cycles	Failures	Cycles	Failures	Cycles
FOR PROTECTION RELIABILITY	0	18	0	23	0	23
	1	40	1	47	1	47

4.4.3.4 List of defects (E5-19-6523 and subsidary drawings). All items contained in this list of defects shall be weighted equally; any defect shall be cause for rejection. Contractor has the option to correct or rework and resubmit for acceptance. When the system is procured in separate subsystems, items 101 through 106 apply to the Shelter and Entrance Assembly with Accessories (PL5-19-6822). and all items except 104, 105 and 106 apply to the M66 Filters and Utilities Unit with Accessories (PL5-19-6822).

Shelter system

101 Painted surfaces not as specified
102 Nameplates, stenciling, and markings not as specified
103 Workmanship (3.5)

Shelter and airlock

104 Caps for inflation connections missing
105 Doors not sealing properly
106 Windows damaged

Air conditioning

107 Coils damaged or leaking
108 Coil fins damaged
109 Compressor misaligned
110 Compressor belt improperly tensioned
111 Controls not properly adjusted
112 Not functioning properly

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:58Z
Check the source to verify that this is the current version before use.

Heater

113 Not functioning properly

Accessories

114 Items missing or improperly stowed

Trailer

u 5 Brakes improperly adjusted, bent, ratchet broken, or missing.
116 Rear support stand improperly installed, missing, bent, or have parts missing (pin).
117 Brake lights not functioning
118 Not properly lubricated

Centrifugal blower

119 Oil level incorrect
120 Blower or drivebelt misaligned
121 Drive belt not properly tensioned

Generator

122 Generator or drivebelt misaligned
123 Drivebelt not properly tensioned
124 Electrical connections not properly mated

Engine

125 Oil level incorrect
126 Motor mounts improperly assembled
127 Evidence of fuel or oil leakage
128 Ignition wiring incorrect
129 Wiring harness improperly routed

Fuel system

130 Pump not properly operating

Electrical system

131 Battery improperly installed, missing, cracked, or damaged
132 Cables improperly routed or connected

Entrance recirculating filter unit

133 Blower rotation incorrect
134 Components missing or incorrectly assembled

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:58Z
Check the source to verify that this is the current version before use.

Filter unit, gas-particulate, entrance

135 Missing or incorrectly installed gaskets
136 Quick release fasteners not aligned
137 Hose clamps loose

Fan assembly, evacuation

138 Airflow direction incorrect
139 Missing components

Auxiliary control panel

140 Press to test feature of lights proper
141 Cool, circulate, heat switch function proper
142 Heater restart function proper
143 High low heat function proper
144 Alarm silence function proper
145 Shelter entrance lights function proper
146 5 minutes and 8 minutes mode function proper
147 Buzzer alarm and mask light function proper

4.4.4 Tests.

4.4.4.1 Initial operation, operating mode, and entry and exit. The M51 shelter system shall be erected and operated in accordance with applicable operating and maintenance instructions contained in TM 3-4240-264-12. The operating mode shall conform to the requirements of 3.3.1. Pressures in the shelter and airlock shall be maintained at values specified in 3.3.2. Entry and exit shall be made at regular intervals in accordance with 3.3.3. The thermostat may be adjusted as required so that the system will operate in heating and cooling modes in accordance with 3.3.1.

4.4.4.2 System leakage. With the M51 fully erected and operational, the entrance and shelter pressures shall be adjusted to a minimum of 0.3 iwg and 0.8 iwg respectively. Door seals and other leak areas shall be tight. The effluent flow shall be measuredly ducting the air through a suitable round duct so that the effluent airflow can be averaged. The minimum allowable flow shall be as specified in 3.3.4.

4.4.4.3 Reliability qualification. The M51 shelter system shall be erected and operated in accordance with applicable instructions contained in TM 3-4240-264-12, Performance requirements of table I shall be measured and recorded. The system shall then be disassembled and prepared for stowage but not stowed,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:58Z
Check the source to verify that this is the current version before use.

4.4.4.4 Production reliability. The M51 shelter systems shall be erected, and operation shall commence in accordance with the applicable instructions contained in TM3-4240-264-12. The systems shall be operated for a period of 1 hour (1/2 hour heating and l/2 hour cooling) with per. formance requirements specified in table I measured and recorded once one half hour into each mode, disassembled, and prepared for stowage but not stowed, Sufficient number of cycles shall be conducted as specified in 4.4,3.3 to demonstrate 0.90 overall system and 0.95 protec. tion reliability at the 60 percent confidence level. Normal operational procedures only shall be allowed during each 1 hour run time per cycle; i.e. no preventive maintenance, component repair or replacement shall be allowed. Any such occurrence necessitating any of the above actions shall be classified" as a protection or system failure as specified in 3.4.3.1 and 3.4.3.2, respectively.

5* PACKAGING

5.1 Preservation, level A. The preservation procedures for the M68 utilities unit and accessories (PL5-19-6821) shall be as follows. The preservation requirements for the shelter and entrance with accessories (PL5-19-6822) shall be in accordance with MIL-S-51422.

5.1.1 Cleaning and drying. The M68 utilities unit and accessories (PL5-l9-6821), less filters, shall be cleaned and dries in accordance with MIL-STD-281.

5.1.2 Preservative application. The M68 utilities unit with accessories (PL5-19-6821) , less filters, shall be preserved in accordance with MIL-STD-281. All system filters identified on Drawing E5-19-5000 and E5-19-6523 shall be preserved in accordance with their applicable specifications (MIL-F-51369 and MIL-F-51366) and packaging data sheets, which are identified by the national stock number.

5.1.3 Unit packing.

5.1.3.1 Accessory items. Accessory items, not permanently mounted, and filter housings and filters shall be packaged as specified herein.

5.1.3.1.1 Tool box contents (D5-19-6525). These tools shall be packaged as specified on D5-19-6525. In the absence of detailed pack-aging description, items susceptible to corrosion shall be packaged in accordance with method I of MIL-P-116 and items not susceptible to corrosion shall be packaged in accordance with method III of MIL-P-116. The packaged contents shall be placed in the tool box as required on D5-19-6525.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:58Z
Check the source to verify that this is the currentversion before use.

5.1.3.1,2 Duct-reducing adapter, duct-12 inch cap, M11 Decontaminating Apparatus, tent pins, log book, technical publications, portable toilet, rod, holding tool handle and driving head (PL5-19-6821). The technical publications (consolidate) and the log book shall be individually packaged method IC-3 of MIL-P-116. The remaining- items except for the M11 Decon and portable toilet shall be individually packaged method III of MIL-P-116. Items supplied in commercial packages and Government furnished items already packaged are not required to be repackaged providing the packaging meets the requirements of packaging method III of MIL-P-116. The M11 Decon (with its holding bracket already removed from the package and assembled on the trailer) and the portable toilet with its accessories shall be removed from their packages and the package materials discarded, Consolidate the items as follows: inclose the duct-reducing adapter, duct-12 inch cap, tent pins, M11 Decon, and portable toilet with its accessories in a weather-resistant, fiberboard box conforming to type CF, variety SW, grade W5c, style RSC of PPP-B-636 with inside dimensions of 20-3/4 by 15-1/2 by 7-1/4 inches (length, width and depth, respectively). The holding tool handle, driving head and rod (attach driving head to rod) shall be packaged together in a weather-resistant fiberboard box ass specified above, except the inside dimensions shall be 37-1/2 by 5 by 5 inches. Cushioning material conforming to type II, class B of PPP-C-843 or PPP-C-1120 shall be included, as necessary in the consolidated packages to prevent movement of the contents. The boxes shall be closed anti strapped in accordance with the closure requirements for weather-resistant boxes in the appendix to PPP-B-636.

5.1.3.1.3 Evacuation manifold assembly, reps'ir kit (inflatable shelter), zipper transition, and pouch and cover assembly (PL5-19-6822). These items shall be packaged in accordance with MIL-S-51422.

5.1.3.1.4 Filter assembly, gas partictiate, (less filters) (E5-19-4917). A wood filler fabricated in accordance with C5-19-6531 shall be secured in place with the fasteners provided.

5.1.3.1.5 Filter unit,gas-partictiate, entrance, (less filters) (E5-19-5009) A wood filler fabricated in accordance with C5-19-5078 shall be secured in place with fasteners provided.

5.1.3.1.6 Filter, gas, (E5-19-5 211-20) and filter, gas (E5-19-5211-10). These filters shall be packaged and marked as specified in MIL-F-51369.

5.1.3.1.7 Filter, particulate, (C5-19-5209-20) and filter, particulate (C5-19-5209-10) . These filters shall be packaged and marked as specified in MIL-F-51366.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:58Z
Check the source to verify that this is the current version before use.

5.1.3.1.8 Filter, gas, 150 CFM, M23 (D5-19-1221) and filter, particulate, 150 CFM, M24 (C5-19-1226). These filters shall be packaged and marked as specified on the packaging data sheet which is identified by the national stock number.

5.2 Packing, level A (see 6.3).

5.2.1 M51 shelter system (less filters) (PL5-19-6523) and accessory items (5.1.3). The accessory items preserved and packaged as specified herein shall be located on the trailer as shown on Drawing E5-19-6523. The shelter system shall be prepared for shipment in accordance with Drawing D5-19-6589.

5.2.2 M66 filter and utilities unit (less filters) (PL5-19-6821) and accessory items (5.1.3.1.1 and 5.1.3.l.2). The accessory items preserved and packaged as specified herein shall be located on the trailer in accordance with Drawing E5-19-6821. The filter and utilities unit shall be prepared for shipment in accordance with Drawing D5-19-6589.

5.2.3 Filter, gas, (E5-19-5211-20) filter, gas, (E5-19-5211-10), filter particulate, (C5-19- 5209-20), filter particulate, (C5-19-5209-10), filter, gas, 150 CFM, M23 (D5-19-1221) and filter particulate, 150 CFM, M24 (C5-19-1226). These filters , packaged as specified in 5.1.3.1.6, 5.1.3.1.7 and 5.1.3.1.8 shall be packed for shipment in a skidded, wood box confomine to class 2 (overseas), style 4, type 1 (easy load), 250-4000 pound weight of contents of PPP-B-621. The packaged filters shall be arranged in the shipping container on end. Fiberboard filler pads conforming to type CF, class weather-resistant, grade W5C of PPP-F-320 shall be included as necessary to make a tight pack. The box shall be closed and strapped in accordance with the appendix to PPP-B-621. The strapping shall conform to type I, class B of QQ-S-781.

5.3 Marking. In addition to any special marking required by the contract or order, marking shall be as follows:

5.3.1 Unit packing (5.1.3). Marking of these items shall be in accordance with MIL-STD-129.

5.3.2 M51 shelter system (less filters) (5.2.1). Marking of the shelter system shall be in accordance with Drawing D5-19-6589.

5.3Q.3 Filters and utilities unit (less filters) (5.2.2). Marking of the filter and utilities unit shall be in accordance with Drawing D5-19-6825.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:58Z
Check the source to verify that this is the current version before use.

5.3.4 Filters (5.2.3). Filters shall be marked for shipment in accordante with MIL-STD-129,

5.3.5 Special marking. Each consolidation container (5.1.3,1,2 and 5.1.3.1.3) shall have pasted to or printed on the upper surface a list of contents, with nomenclature and quantity of each part contained,

5.4 Repair part. When the components of the shelter system (PL5-19-6523) and the filter axial utilities unit (PL5-19-6821) are procured for storage and issue as repair parts , preservation unit and intermediate packing, pecking and marking shall be specified on the packaging data sheets which are identified by the item. national stock number.

6, NOTES

6.1 Intended use. The collective protection equipment covered herein is intended to provide a collective protection system in which the make up ventilation and pressurization air is filtered of toxic gases and aerosols. A protective entrance is provided through which personnel may pass without seriously affecting the pressurization of' the shelter and without introducing contamination into the protected area.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Preproduction and reliability qualifications.

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

c. Type of procurement (see 6.3).

(U Complete system procurement (E5-19-6523).
(2) Split procurement as defined by Drawings PL5-19-6821 and PL5-19-6822.

6.3 This specification covers two types of procurements (1) a complete M51 shelter system consisting of both hardware and software assemblies and accessories (E5-19-6523), and (2) a split procurement in which sybsystem are procured separately. The M68 Filters and Utilities Unit with Accessories is defined by Drawing PL5-19-6821, and the Shelter and Entrance Assembly with Accessories is defined by Drawing PL5-19-6822.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:58Z
Check the source to verify that this is the current version before use.

6.4 International interest. Certain provisions of this specification are the subject of International Standardization Agreement QSTAG-451. When amendment, revision, or cancellation of this specification is proposed which affects or violates the international agreement concened, the preparing activity will take appropriate reconciliation action through international standardization channels including department standardization offices if required.

Custodian:

Army-EA

Preparing activity:

Army-EA

Project No. 4240-A715

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:58Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-37305B

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:58Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DOD-314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Armament Research
and Development Command
ATTN: DRDAR-TSC-S
Aberdeen Proving Ground, MD 21010

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:58Z
Check the source to verify that this is the current version before use.